COUNTABLE INDEXEDNESS TOGETHER WITH CLASSICITY ASCENDS AND DESCENDS ALONG A REPRESENTABLE SURJECTIVE FLAT MORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to the section on taut descent and applies its main statement with the roles of the two rings swapped; that reference is carried as a navigation stub and recorded as a prerequisite.

Editorial note. The counted spans carry 7 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Countable indexedness alone ascends and descends along such a morphism, and the type statements for affine morphisms. Same chapter, Lemma lemma-iff-countably-indexed at proof lines 4604-4606 and Lemma lemma-property-goes-up-affine at proof lines 4602-4604. Gives the first direction and reduces the converse to the case where both spaces are countably indexed.
- Countably indexed affine formal algebraic spaces are formal spectra, and morphisms between formal spectra correspond to continuous algebra homomorphisms. Same chapter, Lemma lemma-countably-indexed at proof lines 4606-4608 and Lemma lemma-morphism-between-formal-spectra at proof lines 4608-4610. Convert the geometric statement into a statement about a ring map.
- A representable affine morphism gives a taut ring map. Same chapter, Lemma lemma-representable-affine, invoked at proof lines 4610-4612. Supplies tautness, which the descent statement requires.
- The fibre product of the spectra is computed by the quotient by the closure of the extended ideal, and the resulting map of quotients is faithfully flat. Same chapter, Lemmas lemma-fibre-product-affines-over-separated and lemma-closure-image-ideal, invoked at proof lines 4614-4619. Identifies the fibre product and shows the map of quotients has the shape needed for the descent statement.
- The descent statement for weakly admissible rings along a taut map of the given shape. Same chapter, Lemma lemma-taut-descent-weakly-admissible, invoked at proof lines 4623-4625. Concludes the lemma.

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Source reference. Original lemma 0AMZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1165–1165. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN1: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2493–2493. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN3: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3703–3703. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4549–4549. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN0: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2355–2355. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4373–4373. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AN5: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4239–4239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXM: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1821–1821. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 0GXK: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1785–1785. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a morphism of affine formal algebraic spaces which is representable by algebraic spaces, surjective, and flat. Then X is countably indexed and classical if and only if Y is countably indexed and classical.*

Proof. We have already seen the implication in one direction in Lemma 0AKQ. For the other direction, note that by Lemma 0AKS we may assume both X and Y are countably indexed. Thus $X = \mathrm{Spf}(A)$ and $Y = \mathrm{Spf}(B)$ for some weakly admissible topological S-algebras A and B, see Lemma 0AN1. By Lemma 0AN0 the morphism

$X \to Y$ corresponds to a continuous S-algebra homomorphism $\varphi : B \to A$. We see from Lemma 0AN5 that φ is taut. Let $J \subset B$ be an open ideal and let $I \subset A$ be the closure of JA. By Lemmas 0AN3 and 0AMZ we see that $\mathrm{Spec}(B/J) \times_Y X = \mathrm{Spec}(A/I)$. Hence $B/J \to A/I$ is faithfully flat (since $X \to Y$ is surjective and flat). This means that $\varphi : B \to A$ is as in Section 0GXK (with the roles of A and B swapped). We conclude that the lemma holds by Lemma 0GXM. □